MATH-1014 T10A

Tutorial 8: Approximate Integration & Polar Coordinates

WANG Yixiao

Department of Mathematics

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Today's Roadmap

1. Approximate Integration
2. Polar Coordinates & Curves

Review: Quadrature Rules

We use numerical integration when finding an antiderivative is impossible (e.g., $\int e^{x^2}\,dx$) or when we only have discrete data points.

Core Formulas ($\Delta x = \frac{b-a}{n}$)

- **Midpoint Rule:** $M_n = \Delta x \sum_{i=1}^{n} f(\overline{x}_i)$, where $\overline{x}_i = \frac{1}{2}(x_{i-1} + x_i)$.
- **Trapezoidal Rule:** $T_n = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)]$.
- **Simpson's Rule (n is even):** Uses parabolic interpolation.
  $S_n = \frac{\Delta x}{3}[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 4f(x_{n-1}) + f(x_n)]$.

Review: Error Bounds

To guarantee accuracy, we use error bounds derived from higher-order derivatives. Let $|f''(x)| \leq k$ and $|f^{(4)}(x)| \leq K$ for $a \leq x \leq b$:

Error Bound Formulas

$$|E_T| \leq \frac{k(b-a)^3}{12n^2}, \qquad |E_M| \leq \frac{k(b-a)^3}{24n^2}$$

$$|E_S| \leq \frac{K(b-a)^5}{180n^4}$$

Remark: The Midpoint error is roughly half the Trapezoidal error. Simpson's Rule converges much faster ($\mathcal{O}(h^4)$) due to higher-order approximations.

Problem 1: Error Bounds & Optimization

Problem 1. Approximate the following integral using **Simpson's Rule**:

$$I = \int_0^1 e^{-x^2}\,dx$$

Find the minimum number of subintervals n required to guarantee an absolute error of less than 10^{-4}.

Hint

You need to bound the 4th derivative $|f^{(4)}(x)| \leq K$ on $[0, 1]$. Remember that n must be an even integer for Simpson's Rule.

Solution 1 (Part 1/2): Finding K

Let $f(x) = e^{-x^2}$. Computing successive derivatives yields:

$$f'(x) = -2xe^{-x^2}$$
$$f''(x) = (4x^2 - 2)e^{-x^2}$$
$$f'''(x) = (12x - 8x^3)e^{-x^2}$$
$$f^{(4)}(x) = (16x^4 - 48x^2 + 12)e^{-x^2}$$

To maximize $|f^{(4)}(x)|$ on $[0, 1]$, let $g(x) = 16x^4 - 48x^2 + 12$. Since $e^{-x^2} \leq 1$ on $[0, 1]$, we analyze $|g(x)|$. $g'(x) = 32x(2x^2 - 3)$, so the only critical point in $[0, 1]$ is $x = 0$.

- At $x = 0$: $|f^{(4)}(0)| = 12 \times 1 = 12$.
- At $x = 1$: $|f^{(4)}(1)| = |-20|e^{-1} \approx 7.36$.

Thus, we can safely take the maximum bound $K = 12$.

Solution 1 (Part 2/2): Solving for n

Using the error bound formula for Simpson's Rule:

$$|E_S| \leq \frac{K(b-a)^5}{180n^4} < 10^{-4}$$

Substitute $K = 12$, $a = 0$, and $b = 1$:

$$\frac{12(1-0)^5}{180n^4} < 10^{-4} \implies \frac{1}{15n^4} < 10^{-4}$$

Solving for n:

$$n^4 > \frac{10000}{15} \approx 666.67 \implies n > \sqrt[4]{666.67} \approx 5.08$$

Since n must be an **even** integer for Simpson's Rule, the minimum required number of subintervals is:

$$\boxed{n = 6}$$

Review: Polar Coordinates & Symmetry

Cartesian to Polar Connection

For a point $P(r, \theta)$, $x = r\cos\theta$ and $y = r\sin\theta$.

Symmetry Tests

- **Polar Axis (x-axis):** Unchanged when $\theta \to -\theta$.
- **The Pole (Origin):** Unchanged when $r \to -r$ or $\theta \to \theta + \pi$.
- **Vertical Line ($\theta = \pi/2$):** Unchanged when $\theta \to \pi - \theta$.

Review: Tangents to Polar Curves

For a polar curve $r = f(\theta)$, $x = f(\theta)\cos\theta$ and $y = f(\theta)\sin\theta$. Using the chain rule, the slope of the tangent line is:

Derivative Formula

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$

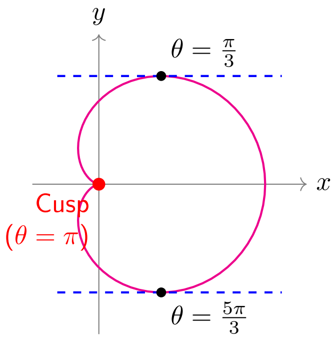
Critical Note: A horizontal tangent occurs when $\frac{dy}{d\theta} = 0$ provided $\frac{dx}{d\theta} \neq 0$. If $\frac{dy}{d\theta} = \frac{dx}{d\theta} = 0$ at the same point, you **must** evaluate $\lim \frac{dy}{dx}$.

Problem 2: Tangent Singularities

Problem 2.
Consider the cardioid given by the polar equation:

$$r = 1 + \cos\theta, \qquad 0 \le \theta \le 2\pi$$

Find all points (r, θ) on the curve where the tangent line is strictly **horizontal**.



Caution

You will encounter a singular pole at $\theta = \pi$ where the curve forms a cusp. Evaluate $\lim_{\theta \to \pi} \frac{dy}{dx}$ using L'Hôpital's rule to determine the true geometric behavior.

Solution 2 (Part 1/2): Finding Critical Points

Express y and x in terms of θ to find their derivatives:

$$y = r\sin\theta = (1+\cos\theta)\sin\theta$$
$$\frac{dy}{d\theta} = -\sin^2\theta + \cos\theta(1+\cos\theta) = 2\cos^2\theta + \cos\theta - 1$$

Setting $\frac{dy}{d\theta} = (2\cos\theta - 1)(\cos\theta + 1) = 0$ yields $\theta \in \left\{\frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$.
Next, we check the denominator $\frac{dx}{d\theta}$:

$$x = r\cos\theta = \cos\theta + \cos^2\theta$$
$$\frac{dx}{d\theta} = -\sin\theta - 2\cos\theta\sin\theta = -\sin\theta(1+2\cos\theta)$$

Solution 2 (Part 2/2): Limit Analysis

At $\theta \in \{\frac{\pi}{3}, \frac{5\pi}{3}\}$, $\frac{dx}{d\theta} \neq 0$. Thus, horizontal tangents occur at:

$$(r, \theta) = \left(\frac{3}{2}, \frac{\pi}{3}\right) \text{ and } \left(\frac{3}{2}, \frac{5\pi}{3}\right)$$

At the cusp $\theta = \pi$, both $\frac{dy}{d\theta} = 0$ and $\frac{dx}{d\theta} = 0$. We evaluate the limit:

$$\lim_{\theta \to \pi} \frac{dy}{dx} = \lim_{\theta \to \pi} \frac{(2\cos\theta - 1)(\cos\theta + 1)}{-\sin\theta(1 + 2\cos\theta)}$$

Since $(2\cos\theta - 1) \to -3$ and $(1 + 2\cos\theta) \to -1$, this simplifies to:

$$3 \lim_{\theta \to \pi} \frac{\cos\theta + 1}{-\sin\theta} \overset{\text{L'H}}{=} 3 \lim_{\theta \to \pi} \frac{-\sin\theta}{-\cos\theta} = 3\left(\frac{0}{1}\right) = 0$$

The limit is 0, confirming a horizontal tangent at $\theta = \pi$.

Final Answer: $(r, \theta) = \left(\frac{3}{2}, \frac{\pi}{3}\right), \left(\frac{3}{2}, \frac{5\pi}{3}\right), (0, \pi)$.

Review: Area and Arc Length in Polar Coordinates

Area of a Polar Region

The area A bounded by the curve $r = f(\theta)$ and rays $\theta = a, \theta = b$ is:

$$A = \int_a^b \frac{1}{2} r^2 \, d\theta = \int_a^b \frac{1}{2} [f(\theta)]^2 \, d\theta$$

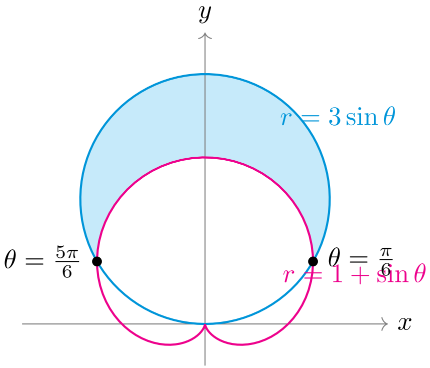
Arc Length of a Polar Curve

The length L of $r = f(\theta)$ for $a \leq \theta \leq b$ is:

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \, d\theta$$

Problem 3: Area Between Polar Curves

Problem 3.
Find the exact area of the region lying **inside** the circle $r = 3\sin\theta$ and **outside** the cardioid $r = 1 + \sin\theta$.



Hint

1. Find the intersection points to establish the limits of integration.
2. Leverage symmetry (calculate the area in Q1 and multiply by 2).
3. Use the area formula $A = \frac{1}{2}\int(r_{outer}^2 - r_{inner}^2)d\theta$.

Solution 3 (Part 1/2): Finding Intersections

Equate the two equations to find the intersection points:

$$3\sin\theta = 1 + \sin\theta \implies 2\sin\theta = 1 \implies \sin\theta = \frac{1}{2}$$

For $\theta \in [0, \pi]$, the intersections are at $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$.
Since both curves are symmetric with respect to the y-axis ($\theta = \pi/2$), we can integrate from $\frac{\pi}{6}$ to $\frac{\pi}{2}$ and double the result:

$$A = 2\int_{\pi/6}^{\pi/2} \frac{1}{2}\left[(3\sin\theta)^2 - (1+\sin\theta)^2\right] d\theta$$

Solution 3 (Part 2/2): Evaluating the Integral

Expand the integrand and apply $8\sin^2\theta = 4(1-\cos 2\theta)$:

$$\begin{aligned}(3\sin\theta)^2 - (1+\sin\theta)^2 &= 8\sin^2\theta - 2\sin\theta - 1 \\ &= (4 - 4\cos 2\theta) - 2\sin\theta - 1 \\ &= 3 - 4\cos 2\theta - 2\sin\theta\end{aligned}$$

Now, evaluate the integral directly:

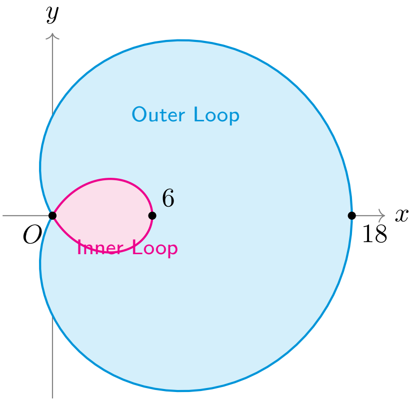
$$\begin{aligned}A &= \int_{\pi/6}^{\pi/2} (3 - 4\cos 2\theta - 2\sin\theta)\, d\theta \\ &= [3\theta - 2\sin 2\theta + 2\cos\theta]_{\pi/6}^{\pi/2}\end{aligned}$$

Substituting the limits ($\pi/2$ and $\pi/6$):

$$A = \left(\frac{3\pi}{2} - 0 + 0\right) - \left(\frac{\pi}{2} - \sqrt{3} + \sqrt{3}\right) = \pi$$

Final Area: $A = \boxed{\pi}$

Problem 4: Area Between Loops of a Limaçon



Find the area between the loops of the limaçon:
$r = 6(1 + 2\cos\theta)$

Integration Setup

The curve passes through the pole ($r = 0$) when $\cos\theta = -1/2$, which gives $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$. **1. Outer Area (Cyan):**

$$A_{\text{out}} = \frac{1}{2}\int_{-\frac{2\pi}{3}}^{\frac{2\pi}{3}} r^2\, d\theta$$

2. Inner Area (Magenta):

$$A_{\text{in}} = \frac{1}{2}\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} r^2\, d\theta$$

Total Area $= A_{\text{out}} - A_{\text{in}}$

Problem 4: Solution Details

First, we expand r^2 using the half-angle identity ($2\cos^2\theta = 1 + \cos 2\theta$):

$$\begin{aligned} r^2 &= 36(1 + 4\cos\theta + 4\cos^2\theta) \\ &= 36(3 + 4\cos\theta + 2\cos 2\theta) \end{aligned}$$

The common antiderivative for our area formula $\frac{1}{2}\int r^2 d\theta$ is:

$$\int \frac{1}{2} r^2 \, d\theta = 18\big(3\theta + 4\sin\theta + \sin 2\theta\big)$$

By utilizing symmetry across the polar axis, we multiply the integral from 0 to the intersection by 2:

$$\begin{aligned} A_{\text{out}} &= 2 \times \frac{1}{2}\int_0^{\frac{2\pi}{3}} r^2 \, d\theta = 18\left[3\theta + 4\sin\theta + \sin 2\theta\right]_0^{\frac{2\pi}{3}} && = 72\pi + 54\sqrt{3} \\ A_{\text{in}} &= 2 \times \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi} r^2 \, d\theta = 18\left[3\theta + 4\sin\theta + \sin 2\theta\right]_{\frac{2\pi}{3}}^{\pi} && = 36\pi - 54\sqrt{3} \end{aligned}$$

$$A_{\text{total}} = A_{\text{out}} - A_{\text{in}} = \mathbf{36\pi + 108\sqrt{3}}$$

THANK YOU!